

Supplementary Material

Protocol-Constrained Residual Adaptation with Frozen Vision–Language Features for Rigid-Body Inference

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This supplement provides implementation details, statistical definitions, dataset-selection information, and instructions for reproducing the saved-result calculations associated with the main article.

1 Claim-to-artifact map

Table 1: Evidence supporting the main article and its interpretation.

Claim	Evidence in the package	Permitted interpretation
C1: 29.95% complete-model improvement	Main result JSON and family action-loss archive; portable paired-statistic replay	Synthetic new-geometry analytic action utility; most gain is numerical.
C2: 1.07% main visual increment	Matched numerical and complete-model family losses	Small selected-feature increment; the two-sided interval includes zero.
C2b: 4.27% public increment	Public error archive and original bootstrap draws	Task-specific fitting followed by offline episode evaluation; not zero-shot or validated object-level transfer.
C3: proper scores and coverage	Main and public result JSON; main cached property means/covariances; selected public radius	Main posterior scores reproduced and mean-fixed sensitivity added; no conformal guarantee.
Capacity: 1,936	Bundled residual checkpoint weight and intercept shapes	Stored residual coefficients only; 1,694 can affect final outputs outside the disabled branch.
Real-video boundary	Saved 33-video, 11-family result JSON	Direct domain inversion and nonsignificant bridge; diagnostic only.

The evidence paths retain their original directory layout under **evidence/**. The accompanying evidence inventory provides file checksums and the retained temporal audit. SHA-256 identifies bytes, not independently verified creation time.

2 Training and label-access sequence

1. Construct observation-only numeric inputs and frozen observation-decoder features. Known force, timing, camera/setup, prior, and linearization inputs are explicit information, not learned visual evidence.

2. For each outer training family partition, fit feature means/scales and the numerical residual heads to anchor-standardized physical targets. Clip the numerical training prediction to two anchor standard deviations.
3. Fit the visual heads on the remaining *in-sample* numerical training residual. Clip their prediction to one standard deviation and multiply by 0.25.
4. Apply both heads to the outer test families, enforce the protocol mask, disable the complete correction for generalized bounce, and calculate accepted-update covariance inflation.
5. Select the final configuration on development data, fit the stored checkpoint, and retain the later confirmation losses separately.

The public extension uses the analogous numerical-then-visual residual, with an additional gate. Four inner folds generate component predictions for gate targets within each of five outer folds. Five candidate gate penalties are scored using those outer predictions. Earlier component selection and gate-penalty selection reuse the 53-episode development cohort. This does not constitute fully nested model-selection validation.

The public radius uses the selected 53 out-of-fold errors at rank 49. Fitting, selection, and residual calibration share the development cohort. The confirmation cohort does not set the radius. We use the term *development-calibrated empirical radial region*, without claiming split-conformal, CV+, or jackknife+ guarantees.

3 Feature and solver implementation details

The main numerical feature dictionary has 24 entries per frame. Nine time-dependent entries are active-axis position, relative position, velocity, velocity difference, log observation noise, time, time increment, time since force switch, and switch indicator. Fifteen repeated setup entries are two forces, one switch time, three active flags, three prior means, three log prior scales, and three prior-normalized linearization coordinates. Flattening eight frames gives 192 inputs. The numerical readout uses no selected visual-residual channel, although it depends on the supplied physical setup and prior specification.

The visual extraction runs the vision component of Qwen3-VL-2B-Instruct, not its text decoder. The model commit is 89644892e4d85e24eaac8bacfd4f463576704203; full model weights are not distributed in this package. Separate image processing uses a 512^2 pixel budget, a 32×32 grid, and BF16 vision evaluation. The frozen observation decoder projects encoder features, combines RGB and tracking, transports temporal state, and pools with spatial attention. The 32 pooled values are states before its static/dynamic correction heads. They are not 32 raw language-model channels. Their temporal mean and normalized static/dynamic position corrections form the 48-dimensional main view.

Relevant source excerpts are:

- `stream_com_internal_features_v16.py`
- `com_observation_decoder_v8.py`
- `train_com_physics_internal_readout_v12.py`
- `sequential_robust_readout_v22.py`

The frozen observation decoder was learned before the residual stage; its parameters must not be hidden inside the 1,936-coefficient claim.

The main robust control computes standardized innovations from a unit-weight linearization, fixes Huber or Cauchy weights, and increases observation noise by the inverse square root of those weights.

It then runs the established nonlinear solver. This is fixed innovation weighting, not a demonstrated converged robust M-estimation loop. The generalized bounce model fixes gravity and allows initial position, velocity, and floor nuisance variables. The setup bounce model fixes release height 0.65 m and zero initial velocity and profiles a free translation. Its code uses three prior-whitened starts $(0, -1, +1)$, three steps per start and three winning-branch steps, finite differences at 10^{-3} , and line-search fractions $(1, 0.5, 0.25)$, totaling 75 forward evaluations per row. Its local curvature covariance omits contact, start-choice, and noise-fit uncertainty. Complete dependency and initialization auditing of the generalized solver is not established by these materials; the excerpts are not a standalone full solver environment.

The public vector numerical head has $(11 + 1)2 = 24$ coefficients, the visual head $(144 + 1)2 = 290$, and the gate $(12 + 1) = 13$, for 327 additional fitted coefficients. In the main checkpoint, the factor 2 counts the generalized and setup conventions, while $2 + 1 + 1$ counts the active scalar outputs for multiple forces, unforced sliding, and bouncing. Each scalar output stores $192 + 1 = 193$ numerical and $48 + 1 = 49$ visual coefficients, including intercepts. Thus $2(2 + 1 + 1)(193 + 49) = 1,936$; 242 in the generalized-bounce branch do not affect the final output, leaving 1,694. They can still be evaluated before the implementation discards their correction, so the active count is not a measured speed reduction.

3.1 Public features and fitted constants

The numerical summary concatenates the final relative centroid (two entries), its last inter-frame difference (two), linear and quadratic coefficients fitted separately to the horizontal and vertical relative coordinates (four), the initial centroid (two), and mean object size (one). Polynomial fits use the eight observation slots at a 0.1 s cadence. The 144-dimensional visual view contains four 32-channel summaries—temporal mean, standard deviation, last-four-minus-first-four mean, and slope against evenly spaced values from -1 to 1 —plus 16 decoder motion channels.

The nine base gate features are tracked endpoint displacement magnitude, dynamic-decoder endpoint displacement magnitude, mean and maximum dynamic-decoder/tracker disagreement, mean dynamic/static disagreement, mean static/tracker disagreement, mean channelwise temporal feature standard deviation, minimum-to-initial object-size ratio, and final-to-initial object-size ratio. Numerical prediction norm, clipped visual correction norm, and their cosine alignment supply the remaining three features. Denominators in the norm/alignment expressions are floored at 10^{-12} ; object-size ratios use 10^{-6} .

All public feature means and standard deviations are fitted on the applicable training partition; feature standard deviations are floored at 10^{-8} . The numerical fit uses a sum-of-squared-residual objective with ridge penalty 10 and an unpenalized intercept, iteratively reweighted by residual vector norms. At each iteration the scale is $\max(1.4826 \text{ median } |r_i - \text{median } r|, 10^{-6})$, where r_i is a residual norm. Weights are $\min(1, 1.345 \text{ scale}/r_i)$. Iteration stops after 50 steps or a maximum weight change below 10^{-8} . The visual and gate fits use ordinary ridge penalties 1,000 and 0.1. The final visual norm limit is 0.067724100878 m. The gate output is clipped to $[0, 1]$; it is not a probability estimate.

The main ridge heads instead use centered targets, a 10^{-6} feature-scale floor, penalties 1 and 10, and coordinate clipping at 2 and 1 anchor standard deviations. The numerical and visual outputs are added with visual shrinkage 0.25. The main and public heads therefore share a sequential residual principle but do not share output targets, hyperparameters, or uncertainty constructions.

4 Statistical replay and cell guards

The bundled script reconstructs the saved main family means and public MAE, RMSE, and median and checks them against result JSON at absolute tolerance 10^{-12} m. It also replays the original main and public paired intervals using stored draws. It does not rerun image extraction, training, physical inference, or posterior generation.

Table 2: Paired differences from saved losses, in millimeters. New fixed-reference and decomposed comparisons are post hoc and unadjusted.

Comparison	Difference	Two-sided 95%	One-sided upper
Main full minus draw-wise strongest	-0.621147	[-0.829331, -0.429618]	-0.459426
Main full minus numerical	-0.015648	[-0.031141, +0.000526]	-0.002189
Main full minus fixed unit	-0.666032	[-0.876075, -0.471932]	-0.503133
Main full minus fixed Huber	-0.621147	[-0.829331, -0.429618]	-0.459426
Main full minus fixed Cauchy	-0.628274	[-0.836708, -0.436415]	-0.466512
Public raw minus numerical	-2.971734	[-5.872120, -0.024801]	-0.511046
Public gated minus raw	-0.195041	[-1.784935, +1.360323]	+1.116326
Public gated minus numerical	-3.166775	[-4.956525, -1.400142]	-1.678076

The primary point improvement is computed from full-precision losses, not rounded table entries. A negative one-sided bound supports direction under the specified procedure; it does not prove a population effect of at least 10%. The public episode intervals assume episode exchangeability; no physical-group clustered interval is presented.

4.1 Descriptive influence sensitivity

For a paired per-unit difference d_i , a deletion mean is $(\sum_j d_j - d_i)/(n - 1)$. These calculations omit one saved evaluation unit at a time without refitting the estimator. They assess concentration of the observed mean, not the sampling distribution of a newly trained model. The following results are post hoc; main units are geometry families and public units are episodes.

Table 3: Paired-unit influence checks. Differences are in millimeters; negative favors the first method. Wins/ties/losses count individual units.

Comparison	Wins/ties/losses	Median difference	Deletion mean range
Main full / Huber	56/0/16	-0.43509	[-0.64891, -0.57501]
Main full / numerical	46/0/26	-0.01117	[-0.01995, -0.01346]
Public gated / numerical	72/5/46	-1.44034	[-3.39173, -2.92134]
Public gated / raw	52/7/64	+1.06307	[-0.50698, +0.07357]

The main visual effect retains a negative mean after any one-family deletion, but its two-sided bootstrap interval still includes zero. The gate-only mean can change sign after one-episode deletion. Neither diagnostic supports increasing the strength of the inferential claims or choosing a new method on the confirmation data. The median paired difference is not the difference between the two marginal medians reported in the main public table.

For completeness, the stored paired draws give post hoc two-sided percentile intervals for relative mean improvement of [22.732, 36.459]% (main full versus fixed Huber), [-0.036, 2.063]% (main visual increment), [1.975, 6.459]% (public gated versus numerical), and [-1.911, 2.525]% (gate alone). Fixed-Huber inference is distinguished from the originally specified draw-wise-strongest comparison. These intervals are unadjusted for multiple comparisons and do not supersede the original analysis rules.

All six meet both the 10% relative and 0.10 mm absolute harm guards. These values preserve the original cell-specific comparator rule; they are not newly estimated cell confidence intervals. Sliding inference is shared by the two conventions, which explains the identical multiple-force and unforced-slide rows.

Table 4: All six point-error protection cells. Both conventions use the same 72 geometry families.

Convention	Protocol	Candidate (mm)	Reference (mm)	Reference
General	Multiple forces	2.989925	3.633112	Huber
General	Unforced slide	0.960083	2.002216	Legacy
General	Bounce	0.515000	0.515000	Huber
Setup	Multiple forces	2.989925	3.633112	Huber
Setup	Unforced slide	0.960083	2.002216	Legacy
Setup	Bounce	0.303715	0.307627	Legacy

5 Public timing, selection, and uncertainty audit

The fixed selection samples a permutation of episode indices 10–1499 with seed 20260924: the first 64 become development candidates and the next 128 confirmation candidates. The local plan and individual selected files are included; a fresh census of the pinned remote population is not included in this analysis.

For every retained episode, the saved video container reports 50 fps. All 176 trajectory parquet files match the stored download hashes. Timestamp differences yield onset-to-target durations from 1.799999833 to 1.800000131 s and last-observation-to-target durations from 1.519999862 to 1.520000100 s. These float32-scale differences support the intended 1.80 s and 1.52 s definitions. A current public-card rate was not substituted for the saved data cadence.

The onset detector uses the first 25 centroids as a baseline, then searches for five consecutive frames displaced by more than two pixels. All frames are tracked before onset selection; later red-component failure can exclude a sequence. Feature extraction remains target-label-free, but selection is not causal with respect to the last observed frame. The result is explicitly scoped to an offline visibility-selected task. A future online result would require a prospectively specified causal selection procedure and new evaluation data.

The nominal 90% region uses radii 0.165462 m for numerical and 0.162973 m for gated prediction. An isotropic Gaussian scale is obtained by dividing the radius by $\sqrt{\chi^2_{2,0.9}}$. This fitted parametric interpretation is used for NLL; it does not transform the empirical radius into a calibrated Gaussian posterior. Object/parameter/environment overlap across episode splits remains unverified.

6 Cached-feature controls and covariance sensitivity

These additional experiments were specified after the original outcomes were available and are descriptive, post hoc, and unadjusted for multiple comparisons. They do not constitute a new independent confirmation and do not select a replacement model. All seven variants and all four covariance scales are reported. Training uses the 132 development families. Their original six folds are reproduced by sorting family identifiers by SHA-256 and assigning consecutive families modulo six; normalization and fitting occur inside each training fold. The 72 previously evaluated families receive models fitted on all development families only. Both cohorts were already available during study design, so the original inference remains primary.

The numerical-only and complete configurations are controls. Pooled-only zeros the 16 normalized position-correction channels; position-only zeros the 32 pooled-state channels. All visual fits retain 48 allocated input dimensions, the same penalty and the same numerical head, although the numbers of nonzero channels differ. These are component controls within the trained observation decoder, not an independent motion tracker or a replacement frozen video encoder. The direct-target variant fits the visual head to the original anchor-standardized target instead of the remaining numerical residual, while retaining numerical addition. The unclipped variant removes both clipping operations during

training-target construction and prediction. The no-fallback variant retains the trained generalized-bounce correction. Removing the coordinate mask alone would be uninformative because the fitted heads already have only active-coordinate outputs; no general mask ablation is claimed.

Table 5: Post hoc cached-feature experiments. Action errors are in millimeters. Development uses six-fold family predictions. Previously evaluated data are descriptive; differences and unadjusted paired 95% intervals are relative to the original full model.

Variant	Development	Evaluated	Difference	95% interval
Numerical only	1.371928	1.468770	+0.01565	$[-0.00053, +0.03114]$
Original full	1.352231	1.453122	reference	—
Pooled states only	1.367787	1.468546	+0.01542	$[+0.00684, +0.02396]$
Position corrections only	1.351850	1.451762	−0.00136	$[-0.01359, +0.00976]$
Direct visual target	1.365456	1.475368	+0.02225	$[-0.00217, +0.04682]$
Unclipped heads	1.364726	1.484064	+0.03094	$[-0.00026, +0.07912]$
No bounce fallback	1.401275	1.494771	+0.04165	$[+0.03078, +0.05253]$

The full and numerical predictions reproduce the original saved outputs and scores within an absolute tolerance of 10^{-12} ; their development scores also match the original six-fold records. Pooled-only provides little point improvement over numerical-only, whereas position corrections retain the measured residual signal. Position-only and full have overlapping paired uncertainty. Direct-target and unclipped variants have higher point errors, but their evaluated-cohort differences from full include zero. Removing generalized-bounce fallback worsens mean action error on both cohorts. These controlled configurations provide evidence about this implementation, without establishing universal safety or a semantic-language contribution.

6.1 Mean-fixed covariance sensitivity

For each scale $\beta \in \{0, 0.05, 0.10, 0.20\}$, the complete model mean is held fixed and $\Sigma_\beta = C + \text{diag}[(\beta d)^2]$. The existing evaluated cohort is rescored without refitting. The original model remains at $\beta = 0.10$; this is descriptive sensitivity rather than confirmation-cohort reselection.

Table 6: Post hoc covariance-scale sensitivity with the complete mean fixed. Width is the mean coordinate-projected joint-region diameter relative to Huber; NLL and CRPS use transformed properties.

Scale	NLL/dim.	CRPS	90% coverage	Width ratio
0.00	-2.532473	0.01310272	0.968056	1.000000
0.05	-2.538692	0.01310534	0.971759	1.000560
0.10	-2.545938	0.01311324	0.973611	1.002236
0.20	-2.543325	0.01314513	0.977315	1.008857

The original scale has the lowest NLL of the four values, whereas zero inflation has the lowest CRPS. Coverage increases with inflation and is already conservative at zero inflation. Thus inflation is not uniformly beneficial across scores and is not responsible for the whole proper-score gain over the robust anchor. Action error is identical at every scale. The width calculation uses $2\sqrt{\chi_{k,0.9}^2 \Sigma_{jj}}$, averaged over active coordinates and the existing family/protocol units; it measures coordinate projections, not principal eigen-axis lengths.

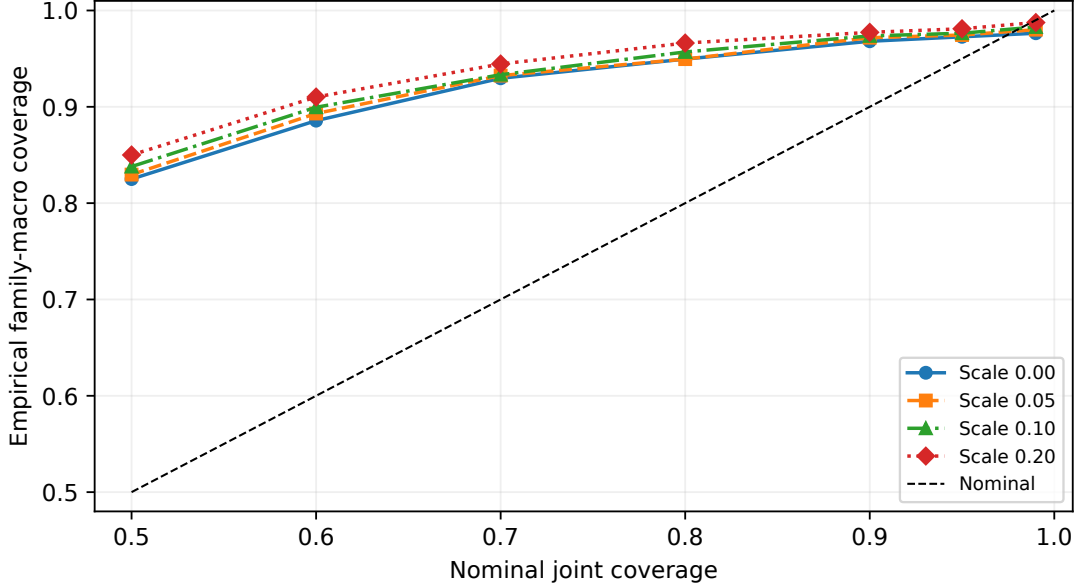


Figure 1: Post hoc empirical coverage at seven nominal levels, recalculated from saved predictions with the complete mean fixed. Symbols and line styles identify covariance scales. The diagonal indicates nominal coverage; no independent calibration sample or confidence bands are introduced.

6.2 Initialization Jacobian audit

The input cache contains the physical derivative matrix J and nuisance derivative matrix J_n at the observation initializer. Residuals are divided by observation noise and physical perturbations are measured in prior-standard-deviation units. We reconstruct $J_{\perp} = J - J_n J_n^+ J$ using the recorded nuisance dimensions and a pseudoinverse cutoff of 10^{-8} . It exactly matches the stored projected derivative in this cache. These are initialization derivatives, not final robust-anchor derivatives or separate setup-bounce derivatives. Singular values below $\max(10^{-10}, 10^{-8}s_{\max})$ count as zero.

Table 7: Post hoc singular-value audit of the active initialization subspace. Condition medians use full-rank rows. All entries are in the cache's standardized coordinate system.

Protocol	Full rank / rows	Median $s_{\min}$	10th pct. $s_{\min}$	Median condition
multiple forces	360/360	0.6557	0.5532	11.3110
unforced slide	360/360	3.6974	3.4170	1.0000
bounce	360/360	21.0018	18.9655	1.0000

The one-dimensional condition number is identically one for a nonzero column and does not itself indicate strong information. Inactive derivative columns are zero by construction in the initializer; this audit therefore cannot establish that the protocol mask equals the null space of an unrestricted physical model. Initial full rank does not remove nonlinear contact ambiguity or guarantee stable posterior inference. In particular, full rank of the bounce initialization does not invalidate the empirically selected final fallback.

7 Independent frozen-video representation control

This additional study was specified after the original results and is post hoc. It uses the official frozen V-JEPA 2 ViT-L encoder, repository `facebook/vjepa2-vitl-fpc64-256`, at the following pinned commit:

`b3c1679b7c34d3255ef3547f27c7b226aefab26f`

All 24,480 original frame files across 1,980 development and 1,080 previously evaluated episodes were hash-checked. Each clip contains exactly the original eight ordered RGB frames. No additional frames, temporal resampling, or physical target is used during feature extraction. Spatial preprocessing uses the encoder’s 256-pixel center crop after its native resize, and encoder evaluation uses BF16. This differs from the Qwen path’s spatial processing.

The final encoder tokens are averaged to 1,024 features. Centered PCA is fitted separately on each training fold and retains 48 components; the evaluation fold never fits the projection. The final projection uses all development features only. The existing numerical head, residual target, penalties, clipping, visual shrinkage, protocol masks, generalized-bounce fallback, and covariance rule are unchanged. The same six development family folds are used. This matches residual-stage rules and output dimension, but not encoder pretraining, observation-decoder supervision, or preprocessing. The PCA map additionally stores 49,152 loadings and 1,024 centering values; these are outside the original residual-coefficient count.

Table 8: Post hoc frozen-video control. Action errors are in millimeters. Development evaluates the residual stage using the original fixed upstream models. The final column gives the existing-cohort paired difference from the original full model and an unadjusted two-sided 95% interval.

Configuration	Development	Evaluated	Difference from full [95% interval]
Numerical only	1.371928	1.468770	+0.01565 [−0.00053, +0.03114]
Original full	1.352231	1.453122	reference
V-JEPA 2, PCA-48	1.382427	1.470615	+0.01749 [−0.00118, +0.03560]

The V-JEPA-minus-numerical difference is +0.00185 mm with interval [−0.00914, +0.01288] mm on the existing evaluation cohort. The full model has the lower point error, but neither this comparison nor V-JEPA versus full establishes a two-sided directional difference. The development V-JEPA-minus-full difference is +0.03020 mm, with interval [+0.01019, +0.05058] mm; it does not supersede the existing-cohort uncertainty. All planned configurations are reported without changing the primary model. V-JEPA’s evaluated NLL/dimension, marginal CRPS, and joint 90% coverage are −2.543389, 0.01318921, and 0.973611, respectively. Cached embeddings, the fixed plan, PCA maps, prediction arrays, and refitting code accompany the package.

8 Upstream lineage and an RGB-path replay

The saved observation-decoder training records identify 328,356 parameters, 720 training episodes, and 48 old-development geometry families. The training objective combines center-of-mass pixel supervision with fixed physical-response sensitivity terms. The frozen encoder is not trained in this stage. An identifier audit finds no episode or family-identifier overlap with the later 72-family confirmation cohort. This is an identifier-level check, not an independent proof of asset or perceptual disjointness. Training receipts and source contracts are supplied; upstream training itself was not rerun.

The residual-development cohort contains these 720 upstream training episodes. Consequently, its six-fold results concern residual fitting conditional on the previously trained decoder, not out-of-fold training of every learned component. This distinction also limits interpretation of the development

comparison with a video encoder that lacks this task-specific decoder supervision. The original later geometry cohort remains the primary evaluation.

The common appearance prior is a separate upstream learned component. Its recorded plan uses 12,000 first-image mappings (2,538 unique image hashes), with 8,400 training and 1,200 validation rows including repeated appearances. A frozen full Qwen pass uses the prompt “Describe only the visible appearance of the object.” The final-token feature undergoes float32 layer normalization, followed by a linear six-output head for three transformed means and three softplus scales. The plan specifies seeds 17, 43, and 101, 100 epochs, and minimum validation Gaussian NLL checkpoint selection; the primary path uses the frozen seed-17 checkpoint. This shared prior is also present in the numerical residual control. These are recorded training specifications, not a fresh reproduction of prior training.

An additional prior-lineage audit checks actual image-file hashes. The 9,600 fitting/selection rows contain 2,036 unique first-image hashes. Against all 1,080 later confirmation episodes (5,703 unique frame hashes), episode identifiers, family identifiers, and byte-exact image hashes have zero overlap. The saved 100-epoch history also reproduces the reported seed-17 selection at epoch 6. This does not prove perceptual or underlying asset independence, or exclude foundation-model pretraining overlap. The identifier/hash tables, training plan, and selection receipts accompany the evidence archive.

The RGB reproduction archive selects the first eight cases of the original observation index without using outcomes, covering all three protocols. Starting from 64 original RGB frames, it regenerates the first-image prior, tracking, numerical observation initialization, Qwen vision and decoder features, robust physical solves, and locked residual predictions. Prior, tracking, initialization, pooled, and position-decoder arrays reproduce exactly in the recorded environment. The maximum transformed-mean discrepancy is 5.29×10^{-10} , and the maximum covariance-entry discrepancy is 6.06×10^{-14} . Before execution, absolute tolerances were set to 10^{-6} for prior outputs, 10^{-4} for numerical inputs and decoder features, and 10^{-5} for posterior entries. Reference arrays are used for assertions; physical targets do not enter prediction.

The archive includes required research modules and small checkpoints, while the pinned Qwen base weights are separately downloaded and hash-checked. An independently extracted copy passes the same assertions while a Python file-read guard rejects dependencies on the original research source/data checkout. It uses the existing Python 3.11/CUDA runtime, not a newly installed environment. To preserve the original prior-head CPU batch shape, eight features are zero-padded to 256 rows; the linear head has no cross-row operation. This checks eight inference paths, not upstream training or full-dataset reproducibility. Run `scripts/run_access_rgb_bundle.py` in the RGB archive with its explicit model directory. Run `tools/score_access_video_control.py` in the Overleaf archive with Python, NumPy, and SciPy to refit the frozen-video control from bundled embeddings; its outputs are written to `replayed/video_control/`.

9 New-episode evaluation with disjoint friction conditions

The follow-up plan was fixed after the original experiments but before downloading new videos or reading new displacement outcomes. It retains the pinned PhysProbe commit. The supplied metadata contain 16 distinct (μ_s, μ_d) pairs for the object and 16 for the surface. These are physical material conditions, not identifiers for distinct cube assets or scenes. An audit finds that the original development/confirmation cohorts share 15 object-material and all 16 surface-material conditions.

All 192 formerly selected episodes and feasibility episodes 0–9 are excluded. Each material list is sorted, then ordered by SHA-256 of the fixed seed, axis name, and material tuple. Seed 2026091307 assigns the first eight materials per axis to training and the remaining eight to evaluation. Episodes must belong to the same role on both axes; the 630 mixed assignments are omitted. This produces 320 training and 348 evaluation candidates. The unchanged full-video visibility procedure retains 285 and 329, excluding 35 and 19 respectively. The package records exclusions by group. The evaluation

contains 63 of 64 possible held-out material combinations. Video hashes have zero overlap across roles. Material values are used for grouping only and are not readout inputs.

The numeric-summary, temporal-visual, and quality-gate views are unchanged. Penalties remain 10, 1,000, and 0.1, respectively; there is no new hyperparameter search. Readout coefficients are newly fitted on the 285 training cases. For every observed training material pair, component models used to form gate targets exclude all rows with either of that pair’s material conditions. The 58 such internal fits cover all training rows. The final numeric and visual components use all training rows, and the gate is fitted from these internally held-out predictions. Original checkpoints and paper results are retained. Architecture and penalty selection historically used other episodes from the same material bank; the separation applies to this refitting experiment, not to every preceding design decision.

Predictions for all 329 evaluation episodes were saved and hashed before any evaluation trajectory was downloaded. Only then were the planar displacement targets read. RGB processing uses the original eight frame offsets and 50 fps horizon; it retains offline full-video eligibility and is not a new causal-online protocol.

9.1 Metric, crossed-material uncertainty, and result

Let $\bar{e}_{ab}^M$ be method M ’s mean vector error in millimeters among eligible episodes with object material a and surface material b . For the 63 observed pairs $\mathcal{C}$, the primary difference is

$$\Delta = |\mathcal{C}|^{-1} \sum_{(a,b) \in \mathcal{C}} (\bar{e}_{ab}^G - \bar{e}_{ab}^N).$$

This gives each observed material pair equal weight. The secondary episode mean instead weights pairs in proportion to their episode counts. They estimate different averages and must not be interchanged when reporting the relative improvement.

For each of 20,000 bootstrap draws, eight object-material labels and eight surface-material labels are independently sampled with replacement. If their multiplicities are w_a and v_b , a draw uses

$$\Delta^* = \frac{\sum_{(a,b) \in \mathcal{C}} w_a v_b (\bar{e}_{ab}^G - \bar{e}_{ab}^N)}{\sum_{(a,b) \in \mathcal{C}} w_a v_b}.$$

The 2.5th and 97.5th percentiles form the reported interval. This applies the pigeonhole resampling principle to the observed material-pair means [S1]. Missing cells are omitted; groups, not individual episodes, determine the crossed resampling weights. This is an approximate interval with only eight levels per axis, not an exact finite-sample guarantee, and it conditions on the fitted models and the visibility-selected data.

Table 9: Additional double-material holdout. The protocol was fixed before these new outcomes; it followed the earlier study. All errors are in millimeters.

Method	Equal material-pair MAE	Episode MAE
Numerical	63.419394	65.091048
Raw visual residual	61.942794	64.641735
Gated visual residual	60.641429	62.753128

The gated-minus-numerical difference is -2.777965 mm, or a 4.38% material-pair reduction, with interval $[-6.043509, -0.115608]$ mm. Marginal mean differences, averaging observed pair means within each material, are negative in seven of eight object-material groups and five of eight surface-material groups. The fixed rule required a negative primary mean, a negative two-sided upper endpoint, and at least five improving groups on each axis; all three conditions pass. The upper endpoint is near zero.

The result supports retention of the visual-plus-gate increment under held-out friction conditions in this refit, with the same cube/scene and previously selected architecture. It does not retroactively pass the original 10% development rule, prove a separate gate-only effect, or establish new object/scene-asset generalization.

Run `tools/replay_group_holdout.py` to refit the models from supplied numeric/visual views, compare the sealed predictions, and replay the crossed-material interval. This uses NumPy and the included original readout routines. It does not re-extract all videos. The full observation plan, video hashes, eligibility records, feature provenance, training/evaluation targets, checkpoints, and predictions are under `evidence/group_holdout/`.

10 Scope of available evidence

The archives identify computational artifacts rather than distinct independent replications. The original public development result remains below its 10% criterion. The controls do not isolate language pretraining, validate a general coordinate mask, or establish universal safety or coverage guarantees. Independent calibration and controlled end-to-end latency comparisons are not available. The new material holdout retains the historical architecture-selection and shared-asset limitations described above.

11 Portable calculation reproduction

The companion source project compiles with pdfLaTeX and BibTeX. The main document is `main.tex`. Select `supplementary.tex` for this supplement or `cover_letter.tex` for the cover letter.

For numerical replay, install NumPy and run these commands outside Overleaf. Outputs are written to `replayed/`.

```
python tools/replay_tables.py
python tools/effect_sensitivity.py
python tools/run_support_study.py
python tools/score_access_video_control.py
python tools/replay_prior_lineage.py
python tools/replay_group_holdout.py
```

The support and video studies additionally require SciPy. Their feature caches and numerical kernels enable CPU refitting and scoring. Input hashes and the analysis plans are supplied under `evidence/support/` and `evidence/video_control/`. The prior-lineage command checks the supplied identifier/hash tables and selection receipts; it does not rehash the complete original image collection.

The separate `IEEE_Access_RGB_Reproduction.zip` includes a README, recorded environment versions, a pinned base-model download helper, and the eight-case offline runner. Its directory structure is independent of this Overleaf project. After obtaining the base model, run:

```
python scripts/run_access_rgb_bundle.py --model-dir /path/to/qwen
```

The checks and discrepancies are written under `reproduced/`. Foundation weights are separate inputs. This example regenerates the first-image prior and executes the physical solver; it does not train upstream components or validate a fresh environment installation.

References

- [S1] A. B. Owen, “The pigeonhole bootstrap,” *The Annals of Applied Statistics*, vol. 1, no. 2, pp. 386–411, 2007, doi: 10.1214/07-AOAS122.